EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is -

(A) 0 (B) 2 (C) 1 (D) 3

MTR001

2. The value of $\sin 10° + \sin 20° + \sin 30° + \ldots + \sin 360°$ is equal to -

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2

MTR002

3. If $\cos x + \cos y + \cos\alpha = 0$ and $\sin x + \sin y + \sin\alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin\alpha$ (B) $\cos\alpha$ (C) $\cot\alpha$ (D) $2\sin\alpha$

MTR003

4. $\frac{\sin(A-C)+2\sin A+\sin(A+C)}{\sin(B-C)+2\sin B+\sin(B+C)}$ is equal to -

(A) $\tan A$ (B) $\frac{\sin A}{\sin B}$ (C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$

MTR004

5. The expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$ is equals -

(A) $\tan\theta$ (B) $\tan 2\theta$ (C) $\sin 2\theta$ (D) $\cos 2\theta$

MTR005

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6. If $\tan\alpha = (1 + 2^{-x})^{-1}$, $\tan\beta = (1 + 2^{x+1})^{-1}$, then $\alpha + \beta =$

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

MTR009

7. If $\tan B = \frac{n\sin A\cos A}{1 - n\cos^2 A}$ then $\tan(A + B)$ equals

(A) $\frac{\sin A}{(1-n)\cos A}$ (B) $\frac{(n-1)\cos A}{\sin A}$ (C) $\frac{\sin A}{(n-1)\cos A}$ (D) $\frac{\sin A}{(n+1)\cos A}$

MTR010

8. $\frac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta} =$

(A) $\frac{1}{2}\tan\theta$ (B) $\frac{1}{2}\cot\theta$ (C) $\tan\theta$ (D) $\cot\theta$

MTR006

9. If $A = \tan 6° \tan 42°$ and $B = \cot 66° \cot 78°$, then -

(A) $A = 2B$ (B) $A = 1/3B$ (C) $A = B$ (D) $3A = 2B$

MTR007

10. In a right angled triangle, the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are

(A) $\frac{\pi}{3}$ and $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ and $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ and $\frac{3\pi}{10}$

MTR008

For More Join: @IITJEE_Advanced

11. The value of $\operatorname{cosec}\frac{\pi}{18}-\sqrt{3}\sec\frac{\pi}{18}$ is a

(A) surd
(B) rational which is not integral
(C) negative integer
(D) natural number

MTR011

12. The value of $\cot x+\cot(60^\circ+x)+\cot(120^\circ+x)$ is equal to :

(A) $\cot 3x$ (B) $\tan 3x$ (C) $3\tan 3x$ (D) $\frac{3-9\tan^2 x}{3\tan x-\tan^3 x}$

MTR012

13. If $\frac{5\pi}{2}<x<3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$ (B) $\cot\frac{x}{2}$ (C) $\tan\frac{x}{2}$ (D) $-\tan\frac{x}{2}$

MTR013

14. The value of $\cot 7\frac{1}{2}^\circ+\tan 67\frac{1}{2}^\circ-\cot 67\frac{1}{2}^\circ-\tan 7\frac{1}{2}^\circ$ is :

(A) a rational number
(B) irrational number
(C) $2(3+2\sqrt{3})$
(D) $2(3-\sqrt{3})$

MTR014

15. If $x+y=3-\cos 4\theta$ and $x-y=4\sin 2\theta$ then

(A) $x^4+y^4=9$ (B) $\sqrt{x}+\sqrt{y}=16$
(C) $x^3+y^3=2(x^2+y^2)$ (D) $\sqrt{x}+\sqrt{y}=2$

MTR015

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16. If $\tan A+\tan B+\tan C=\tan A.\tan B.\tan C$, then -

(A) A, B, C must be angles of a triangle
(B) the sum of any two of A, B, C is equal to the third
(C) $A+B+C$ must be an integral multiple of π
(D) None of these

MTR016

17. If $f(x)=\frac{\sin 3x}{\sin x}$, $x\neq n\pi$, then the range of values of $f(x)$ for real values of x is -

(A) $[-1,3]$ (B) $(-\infty,-1]$ (C) $(3,+\infty)$ (D) $[-1,3)$

MTR017

18. Maximum and minimum value of $2\sin^2\theta-3\sin\theta+2$ is -

(A) $\frac{1}{4},-\frac{7}{4}$ (B) $\frac{1}{4},\frac{21}{4}$ (C) $\frac{21}{4},-\frac{3}{4}$ (D) $7,\frac{7}{8}$

MTR018

19. For $\theta\in(0,\pi/2)$, the maximum value of $\sin\left(\theta+\frac{\pi}{6}\right)+\cos\left(\theta+\frac{\pi}{6}\right)$ is attained at $\theta=$

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

MTR019

20. Minimum value of the expression $\cos^2\theta-(6\sin\theta\cos\theta)+3\sin^2\theta+2$, is -

(A) $4+\sqrt{10}$ (B) $4-\sqrt{10}$ (C) 0 (D) 4

MTR020

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $\cos x = \tan x$, then which of the following is/are true?

(A) $\frac{1}{\sin x} + \cos^4 x = 1$ (B) $\frac{1}{\sin x} + \cos^4 x = 2$

(C) $\cos^4 x + \cos^2 x = 1$ (D) $\cos^4 x + \cos^2 x = 2$

MTR045

2. If $\cos(A-B) = \frac{3}{5}$ and $\tan A \tan B = 2$, then which of the following is/are correct ?

(A) $\cos A \cos B = -\frac{1}{5}$ (B) $\sin A \sin B = \frac{2}{5}$ (C) $\cos(A+B) = -\frac{1}{5}$ (D) $\sin A \cos B = \frac{4}{5}$

MTR046

3. If $3\sin\beta = \sin(2\alpha + \beta)$, then $\tan(\alpha + \beta) - 2\tan\alpha$ is

(A) independent of α

(B) independent of β

(C) dependent of both α and β

(D) independent of α but dependent of β

MTR048

4. If $x + y = z$, then $\cos^2 x + \cos^2 y + \cos^2 z - 2\cos x \cos y \cos z$ is equal to

(A) $\cos^2 z$ (B) $\sin^2 z$ (C) $\cos(x + y - z)$ (D) 1

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MTR051

5. If $L = \cos^2 84° + \cos^2 36° + \cos 36° \cos 84°$

$M = \cot 73° \cot 47° \cot 13°$

$N = 4\sin 156° \sin 84° \sin 36°$, then which of the following option(s) is(are) correct ?

(A) $L < 1$ (B) $M > \tan 2$ (C) $N > \sin\frac{\pi}{4}$ (D) $0 < LMN$

MTR049

6. If $P_n = \cos^n\theta + \sin^n\theta$ and $Q_n = \cos^n\theta - \sin^n\theta$, then which of the following is/are true.

(A) $P_n - P_{n-2} = -\sin^2\theta \cos^2\theta \, P_{n-4}$ (B) $Q_n - Q_{n-2} = -\sin^2\theta \cos^2\theta \, Q_{n-4}$

(C) $P_4 = 1 - 2\sin^2\theta \cos^2\theta$ (D) $Q_4 = \cos^2\theta - \sin^2\theta$

MTR054

7. If $\alpha + \beta + \gamma = 2\pi$, then

(A) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(B) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} + \tan\frac{\beta}{2}\tan\frac{\gamma}{2} + \tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 1$

(C) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = -\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(D) $\tan\frac{\alpha}{4}\tan\frac{\beta}{4} + \tan\frac{\beta}{4}\tan\frac{\gamma}{4} + \tan\frac{\gamma}{4}\tan\frac{\alpha}{4} = 1$

MTR050

For More Join: @IITJEE_Advanced

8. Let $y = \dfrac{\cos x+\cos 2x+\cos 3x+\cos 4x+\cos 5x+\cos 6x+\cos 7x}{\sin x+\sin 2x+\sin 3x+\sin 4x+\sin 5x+\sin 6x+\sin 7x}$, then which of the following hold good?

(A) The value of y when $x = \pi/8$ is not defined.

(B) The value of y when $x = \pi/16$ is 1.

(C) The value of y when $x = \pi/32$ is $\sqrt{2}-1$.

(D) The value of y when $x = \pi/48$ is $2+\sqrt{3}$.

MTR053

9. Let $f_n(\theta)=\sum_{r=0}^{n}\frac{1}{4^r}\sin^4\left(2^r\theta\right)$. Then which of the following alternative(s) is/are correct ?

(A) $f_2\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$ (B) $f_3\left(\frac{\pi}{8}\right)=\frac{2+\sqrt{2}}{4}$ (C) $f_4\left(\frac{3\pi}{2}\right)=1$ (D) $f_5(\pi) = 0$

MTR052

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 10 to 12

Consider the polynomial $P(x) = (x - \cos 36°)(x - \cos 84°)(x - \cos 156°)$

10. The coefficient of x^2 is

(A) 0 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{5}-1}{2}$

MTR055

11. The coefficient of x is

(A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{3}{4}$ (D) zero

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MTR056

12. The absolute term in $P(x)$ has the value equal to

(A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}-1}{16}$ (C) $\frac{\sqrt{5}+1}{16}$ (D) $\frac{1}{16}$

MTR057

MATCHING LIST TYPE QUESTION

13.

	List-I		**List-II**
(I)	The value of $\tan 20°\cdot \tan 40°\cdot \tan 60°\cdot \tan 80°$ is	(P)	Odd natural Number
(II)	The value of $4\left(\sin^4\frac{\pi}{16}+\sin^4\frac{3\pi}{16}+\sin^4\frac{5\pi}{16}+\sin^4\frac{7\pi}{16}\right)$ is	(Q)	Even natural Number
(III)	$\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \lambda\cot\lambda\alpha$ then λ is	(R)	3
(IV)	$(4\cos^2 9° - 3)\,(4\cos^2 27° - 3) = \tan k°$, then k is	(S)	6
		(T)	1
		(U)	9

Which of the following is **CORRECT** ?

(A) (I) → (P,R) ; (II) → (Q,S) ; (III) → (P,T) ; (IV) → (P, U)

(B) (I) → (P,T) ; (II) → (Q,R) ; (III) → (P,U) ; (IV) → (R, U)

(C) (I) → (R,S) ; (II) → (T,U) ; (III) → (P,T) ; (IV) → (P, Q)

(D) (I) → (P,R) ; (II) → (P,U) ; (III) → (Q,S) ; (IV) → (S, T)

MTR058

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

14. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities.

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

	Column-I		**Column-II**
(A)	The value of $\left\|4\left(2\cos^3\frac{\pi}{7}-\cos^2\frac{\pi}{7}-\cos\frac{\pi}{7}\right)\right\|$ is	(P)	4
(B)	If $A+B+C=\pi$ and $\cos A=\cos B\cdot\cos C$ then $\tan B\cdot\tan C$ has the value equal to	(Q)	8
(C)	$4\left(\frac{\cos 20^\circ+8\sin 70^\circ\sin 50^\circ\sin 10^\circ}{\sin^2 80^\circ}\right)$ is equal to	(R)	2
(D)	The maximum value of $12\sin\theta-9\sin^2\theta$ is	(S)	1
		(T)	6

MTR059

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For More Join: @IITJEE_Advanced

EXERCISE (S)

1. $2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta) + 1$ is equal to -

MTR021

2. If $x = y\cos\frac{2\pi}{3} = = z\cos\frac{4\pi}{3}$ then $xy + yz + zx =$

MTR022

3. If $\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$ and $2010\tan\beta + 1 = 0$, then $|\tan\alpha|$ is equal to

MTR023

4. Calculate: $\frac{2\cos 40° - \cos 20°}{\sin 20°}$

MTR064

5. If acute angle $A = 3B$ and $\sin A = \frac{4}{5}$ then $\left|\frac{3\sec B - 4\operatorname{cosec} B}{2}\right|$ is

MTR060

6. If $3\sin\alpha = 5\sin\beta$, then $\frac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}} =$

MTR025

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7. Given that $(1 + \tan1°)(1 + \tan2°)\ldots(1 + \tan45°) = 2^n$, find $\left(\frac{n}{4}\right)$.

MTR063

8. If the value of the expression $\sin25°\cdot\sin35°\cdot\sin85°$ can be expressed as $\frac{\sqrt{a}+\sqrt{b}}{c}$, where $a, b, c \in N$ and are in their lowest form, find the value of $\left(\frac{a+b+c}{18}\right)$.

MTR062

9. If $\tan x + \tan y = 25$ and $\cot x + \cot y = 30$, then the value of $\tan(x + y)$ is

MTR024

10. If $\cos\alpha = \frac{2\cos\beta - 1}{2 - \cos\beta}$ then $\tan^2\frac{\alpha}{2}\cdot\cot^2\frac{\beta}{2}$ has the value equal to {where $\alpha, \beta \in (0, \pi)$}

MTR026

11. If the value of $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$ is $\frac{p}{q}$ (in simplest form) then what is the value of $p + q$?

MTR028

12. The exact value of $\frac{96\sin80°\sin65°\sin35°}{\sin20° + \sin50° + \sin110°}$ is equal to

MTR029

For More Join: @IITJEE_Advanced

13. If m and n are positive integers satisfying

$1+\cos 2\theta+\cos 4\theta+\cos 6\theta+\cos 8\theta+\cos 10\theta=\dfrac{\cos m\theta\cdot\sin n\theta}{\sin\theta}$ then $(m+n)$ is equal to

MTR030

14. If $A=\sin\dfrac{2\pi}{7}+\sin\dfrac{4\pi}{7}+\sin\dfrac{8\pi}{7}$ and $B=\cos\dfrac{2\pi}{7}+\cos\dfrac{4\pi}{7}+\cos\dfrac{8\pi}{7}$ then A^2+B^2 is equal to

MTR027

15. If $k_1=\tan 27\theta-\tan\theta$ and $k_2=\dfrac{\sin\theta}{\cos 3\theta}+\dfrac{\sin 3\theta}{\cos 9\theta}+\dfrac{\sin 9\theta}{\cos 27\theta}$ then $\left(\dfrac{k_1}{k_2}\right)$ is equal to

MTR061

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, the expression $3(\sin\theta - \cos\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$ equals : **[JEE (Main) 2019]**

(1) $13 - 4\cos^6\theta$ (2) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

(3) $13 - 4\cos^2\theta + 6\cos^4\theta$ (4) $13 - 4\cos^2\theta + 6\sin^2\theta cos^2\theta$

MTR034

2. The value of $\cos\frac{\pi}{2^2}\cdot\cos\frac{\pi}{2^3}\cdot.....\cdot\cos\frac{\pi}{2^{10}}\cdot\sin\frac{\pi}{2^{10}}$ is : **[JEE (Main) 2019]**

(1) $\frac{1}{256}$ (2) $\frac{1}{2}$ (3) $\frac{1}{512}$ (4) $\frac{1}{1024}$

MTR035

3. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for $k = 1,2,3, \ldots$ Then for all $x \in R$, the value of $f_4(x) - f_6(x)$ is equal to :- **[JEE (Main) 2019]**

(1) $\frac{5}{12}$ (2) $\frac{-1}{12}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

MTR036

4. The maximum value of $3\cos\theta + 5\sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is : **[JEE (Main) 2019]**

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(1) $\sqrt{19}$ (2) $\frac{\sqrt{79}}{2}$ (3) $\sqrt{31}$ (4) $\sqrt{34}$

MTR037

5. Let α and β be two real roots of the equation $(k+1)\tan^2 x - \sqrt{2}\lambda\tan x = (1-k)$, where $k \neq -1$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is ; **[JEE (Main) 2020]**

(1) 5 (2) 10 (3) $5\sqrt{2}$ (4) $10\sqrt{2}$

MTR038

6. The value of $\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right)$ is : **[JEE (Main) 2020]**

(1) $\frac{1}{4}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{1}{2}$

MTR039

7. If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\text{cosec}^6\alpha$ is equal to : **[JEE (Main) 2021]**

(1) 350 (2) 500 (3) 400 (4) 250

MTR040

8. $16\sin(20°)\sin(40°)\sin(80°)$ is equal to: **[JEE (Main) 2022]**

(1) $\sqrt{3}$ (2) $2\sqrt{3}$ (3) 3 (4) $4\sqrt{3}$

MTR041

For More Join: @IITJEE_Advanced

9. If $\cot\alpha = 1$ and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha+\beta)$ and the quadrant in which $\alpha+\beta$ lies, respectively are **[JEE (Main) 2022]**

(1) $-\frac{1}{7}$ and IVth quadrant (2) 7 and Ist quadrant

(3) – 7 and IVth quadrant (4) $\frac{1}{7}$ and Ist quadrant

MTR042

10. The value of $\cos\left(\frac{2\pi}{7}\right)+\cos\left(\frac{4\pi}{7}\right)+\cos\left(\frac{6\pi}{7}\right)$ is equal to : **[JEE (Main) 2022]**

(1) –1 (2) $-\frac{1}{2}$ (3) $-\frac{1}{3}$ (4) $-\frac{1}{4}$

MTR043

11. If $\sin^2(10°)\sin(20°)\sin(40°)\sin(50°)\sin(70°) = \alpha - \frac{1}{16}\sin(10°)$, then $16 + \alpha^{-1}$ is equal to _____.

[JEE (Main) 2022]

MTR044

12. The value of $36\,(4\cos^2 9° - 1)(4\cos^2 27° - 1)(4\cos^2 81° - 1)(4\cos^2 243° - 1)$ is **[JEE (Main) 2023]**

(1) 54 (2) 18 (3) 27 (4) 36

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MTR074

13. If $\tan 15° + \frac{1}{\tan 75°} + \frac{1}{\tan 105°} + \tan 195° = 2a$, then the value of $\left(a+\frac{1}{a}\right)$ is : **[JEE (Main) 2023]**

(1) 4 (2) $4-2\sqrt{3}$ (3) 2 (4) $5-\frac{3}{2}\sqrt{3}$

MTR075

14. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to **[JEE (Main) 2023]**

(1) 3 (2) 2 (3) 4 (4) 1

MTR076

15. Let $f: \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\left\{\sqrt{2}(\sin x - \cos x) + m - 2\right\}$, for some m, such that the range of f is [0, 2]. Then the value of m is. **[JEE (Main) 2023]**

(1) 5 (2) 3 (3) 2 (4) 4

MTR077

16. Let the set of all $a \in R$ such that the equation $\cos 2x + a\sin x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9° - \tan 27° - \frac{1}{\cot 63°} + \tan 81°$, then pqr is equal to ____. **[JEE (Main) 2024]**

MTR126

For More Join: @IITJEE_Advanced

17. For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha+\beta)=2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to : **[JEE (Main) 2024]**

(1) $-\frac{2}{3}$ (2) –5 (3) $\frac{2}{3}$ (4) 5

MTR127

18. Let a rectangle $ABCD$ of sides 2 and 4 be inscribed in another rectangle $PQRS$ such that the vertices of the rectangle $ABCD$ lie on the sides of the rectangle $PQRS$. Let a and b be the sides of the rectangle $PQRS$ when its area is maximum. Then $(a + b)^2$ is equal to : **[JEE (Main) 2024]**

(1) 72 (2) 60 (3) 80 (4) 64

MTR128

19. If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}, \tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C = \left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}, 0 < A, B, C < \frac{\pi}{2}$, then $A + B$ is equal to : **[JEE (Main) 2024]**

(1) C (2) $\pi - C$ (3) $2\pi - C$ (4) $\frac{\pi}{2} - C$

MTR129

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then - **[JEE (Advanced) 2006]**

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$

(C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

MTR065

2. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then **[JEE (Advanced) 2009]**

(A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$ (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

MTR066

3. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \text{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \text{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) - **[JEE (Advanced) 2009]**

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

MTR067

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4. The maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$ is **[JEE (Advanced) 2010]**

MTR068

5. Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ b be two sets. Then **[JEE (Advanced) 2013]**

(A) $P \subset Q$ and $Q - P \neq \varnothing$ (B) $Q \not\subset P$

(C) $P \not\subset Q$ (D) $P = Q$

MTR070

6. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to **[JEE (Advanced) 2016]**

(A) $3 - \sqrt{3}$ (B) $2(3 - \sqrt{3})$ (C) $2(\sqrt{3} - 1)$ (D) $2(2 + \sqrt{3})$

MTR071

7. Let α and β be nonzero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true? (where $\alpha, \beta \neq (2k+1)\pi$) **[JEE (Advanced) 2017]**

(A) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$ (B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(C) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$ (D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

MTR072

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8. Let α and β be real numbers such that $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$. If $\sin(\alpha+\beta)=\frac{1}{3}$ and $\cos(\alpha-\beta)=\frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$ is _____.

[JEE (Advanced) 2022]

MTR073

9. Let $\frac{\pi}{2} < x < \pi$ be such that $\cot x = \frac{-5}{\sqrt{11}}$. Then

$\left(\sin\frac{11x}{2}\right)(\sin6x - \cos6x) + \left(\cos\frac{11x}{2}\right)(\sin6x + \cos6x)$

is equal to :

[JEE (Advanced) 2024]

(A) $\frac{\sqrt{11}-1}{2\sqrt{3}}$ (B) $\frac{\sqrt{11}+1}{2\sqrt{3}}$ (C) $\frac{\sqrt{11}+1}{3\sqrt{2}}$ (D) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

MTR130

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	B	B	A	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	D	B	D	C	D	D	A	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	B,C	A,B	C,D	A,B,C,D	A,B,C,D	A,D	B,D	C,D	A
Que.	11	12	13	14						
Ans.	C	B	A	A→S; B→R; C→Q; D→P						

EXERCISE (S)

1.	0	**2.**	0	**3.**	1	**4.**	1.73	**5.**	5
6.	4	**7.**	5.75	**8.**	1.33	**9.**	150	**10.**	3
11.	9	**12.**	24	**13.**	11	**14.**	2	**15.**	2

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	4	1	2	3	4	2	1	2
Que.	11	12	13	14	15	16	17	18	19	
Ans.	80	4	1	1	1	48	2	1	1	

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	A,B	C,D	2	D	C	A,C	1	B	

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Important Notes

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